Subject: FW: NSF IDSS Proposal Concept
Date: Thursday, May 28, 2026 at 4:21:10 PM Eastern Daylight Time
From: David Molik
To: Gwen Sibley

More K-FORGE info

--
David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/

Book time to meet with me

From: David Molik <molik@ksu.edu>
Date: Monday, May 18, 2026 at 11:05 AM
To: Nicholas Eggleston <njeggleston@ksu.edu>; Jeffrey Sheldon <jsheldon@ksu.edu>
Cc: Adrian Self <amself@ksu.edu>; Mickey Slimp <mick@greatplains.net>; Susan Metzger <smetzger@ksu.edu>; Ryan Rafferty <rjraff@ksu.edu>; Daniel Andresen <dan@ksu.edu>; Kim Rewinkel <krewinkel@ksu.edu>; Nidhi Bhandari <nidhim@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

Hi Nick and Jeff,

I'm starting to shape the budget for the IDSS Category II proposal and wanted to share the general funding pattern I have in mind before I get too far into Cayuse.

I'm currently using **K-FORGE** as the working project name, with the likely full title:

K-FORGE: A Federated Open Research Gateway for Executable, Reproducible, and AI-Enabled Science

Also, I might have used a different name on Cayuse, I'll fix that now. Given Beocat's existing cluster naming style: Heroes, Dwarves, Moles, Wizards, and Warlocks, K-FORGE seems like a good fit as a grant acronym: it nods to the existing Beocat culture while still giving us a serious NSF-facing frame around building, hardening, registering, and executing reusable research environments.

The basic proposal frame is that K-FORGE would build a shared research computing and open research infrastructure layer for K-State, scaled toward national use. The main thrust of the grant is still **containers, container publishing, reproducible workflow execution, metadata, citation, public access, and service integration**. The hardware should support that story, not overtake it.

At a high level, I'm thinking about the budget in these categories:

Category	General purpose
PI/Co-PI effort	Project leadership, technical direction, governance, and integration
Research software engineering / DevOps	Core build capacity for registry, workflow execution, automation, deployment, and service reliability
Project/service coordination	Onboarding, documentation, scheduling, user support, reporting, and coordination across OPE, Beocat, and Sunderland
Beocat/OPE/Sunderland technical support	Integration labor across HPC/Kubernetes execution, repository/public-access systems, and AI Studio workflow translation
Student / GRA / hourly support, including Sunderland students	Workflow testing, documentation, pilots, data preparation, AI Studio prototyping, demonstrations, and support tasks
Modest on-prem infrastructure	Emerging-architecture testbed, local storage/cache, networking, rack hardware, and related components
Software/security/monitoring/backups	Operational reliability, monitoring, security review, backup support, and service hardening
Consultants/evaluation/security review	Targeted outside review or evaluation where useful
Participant support / onboarding / workshops	User onboarding, pilot community engagement, and training
Travel	Limited travel for NSF/community coordination and dissemination
Other direct costs	Miscellaneous project needs that do not fit cleanly elsewhere

For the infrastructure piece, I'm thinking we should be explicit that **edge-AI is the niche**, not GPU-cluster competition. We are not going to compete with conventional GPU clusters, and we should not pretend that we are. The more interesting angle is that edge-AI workflows are a relatively unexplored space for reproducible research infrastructure: small ARM systems, local storage/cache, accelerator modules such as FPGA/TPU/NPU-class devices, and containerized workflows that need to be published, cited, redeployed, benchmarked, and compared across heterogeneous hardware.

Some of this would live in Beocat as a managed experimental environment, and some would support Sunderland/AI Studio development, demonstration, and workflow

translation. I also think Sunderland is especially well positioned for the student-facing side of this work: student prototyping, hands-on testing, demonstrations, documentation, and translation of workflows before they are hardened for Beocat-facing execution. Those student roles may or may not need to appear as a separate line item, but I want to make sure we explicitly plan for Sunderland student involvement in the work.

I also think we may be able to sneak in a few storage/cache servers as part of the testbed infrastructure. I do not know what Beocat currently uses for distributed storage, if it is something like Gluster, Ceph, or another clustered storage model, these could potentially mostly function as storage/cache servers for the test bench rather than general-purpose production storage. The idea would be to support container layers, datasets, model weights, intermediate outputs, benchmark artifacts, and repeated workflow staging without claiming that we are solving Beocat-wide storage needs.

The overall budget philosophy is still salary/service-development first. The hardware should be meaningful enough to make the workflows real, but not so large that the project reads as a hardware acquisition proposal. In other words, the edge-AI testbed is there to support the larger K-FORGE argument: reproducible, citable, containerized research environments that can be published, executed, preserved, and reused across different research and hardware contexts.

I'd appreciate your reactions on whether these are generally the right things to fund and whether anything important is missing before I start tightening the budget.

Best,
David

--
David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/
Book time to meet with me

From: David Molik <molik@ksu.edu>
Date: Monday, May 11, 2026 at 11:03 AM
To: Nicholas Eggleston <njeggleston@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>; Mickey Slimp <mick@greatplains.net>
Cc: Adrian Self <amself@ksu.edu>; Jeffrey Sheldon <jsheldon@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

Morning all,

Mickey Slimp, director of the Great Plains Network has agreed to take the External advisor role for the IDSS application.

I've been thinking about Ryan Rafferty and possibly Susan Metzger as the other two on the advisory board, with no objections I'll reach out today.

--
David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/

Book time to meet with me

From: Nicholas Eggleston <njeggleston@ksu.edu>
Date: Wednesday, May 6, 2026 at 8:51 AM
To: David Molik <molik@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Adrian Self <amself@ksu.edu>; Jeffrey Sheldon <jsheldon@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

After chatting with Dan, seems like he's in favor of these assignments and he's noodling on people who might fill the other roles well.

--
Nick Eggleston
Beocat Team Lead
High Performance Computing Architect
Institute for Computational Research in Engineering and Science

---

From: David Molik <molik@ksu.edu>
Sent: Tuesday, May 5, 2026 11:00 AM
To: Nicholas Eggleston <njeggleston@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Adrian Self <amself@ksu.edu>; Jeffrey Sheldon <jsheldon@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

At the moment, I am thinking about the grant leadership structure this way, but we can change this as needed:

Core leadership

PI: David Molik, PhD
Overall project frame; OPE/repository/public-access/research object layer;

governance, sustainability, and integration across the deliverables.

Co-PI: Nicholas Eggleston
Beocat execution layer; Kubernetes/HPC integration; operational feasibility; infrastructure risk and mitigation.

Co-PI: Jeffrey Sheldon
Sunderland Innovation Lab / AI Studio; emerging technology testbed; user-facing training, demonstration, and workflow translation.

Scientific/technical advisors

Daniel Andresen, PhD
CSE/HPC/scientific computing architecture review, especially if his role is advisory rather than ownership of a full technical work package.

Adrian Self
Research security, cyberbiosecurity, compliance, operational risk, and sensitive-data/governance perspective.

I also think we should consider adding roughly three additional advisors, each serving a different purpose:

1. ***External CI / research computing advisor***: someone who can speak to shared cyberinfrastructure, institutional computing services, and national-scale operations.
2. ***ID3A-affiliated domain scientist***: someone with data-intensive or AI-heavy research needs who can validate campus demand and provide a pilot use case.
3. ***Second domain scientist from a different workflow community***: ideally someone from agriculture/genomics, geospatial/environmental data, biosecurity/cyberbiosecurity, or another community that helps demonstrate cross-domain reuse.

--
David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/
Book time to meet with me

From: Nicholas Eggleston <njeggleston@ksu.edu>
Date: Tuesday, May 5, 2026 at 9:19 AM
To: David Molik <molik@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen

<dan@ksu.edu>
Cc: Gina Becker <gbecker@ksu.edu>; Adrian Self <amself@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

Dan and I are meeting this afternoon and we can discuss it and get back to you.

--

Nick Eggleston
Beocat Team Lead
High Performance Computing Architect
Institute for Computational Research in Engineering and Science

---

From: David Molik <molik@ksu.edu>
Sent: Tuesday, May 5, 2026 9:18 AM
To: Nicholas Eggleston <njeggleston@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Gina Becker <gbecker@ksu.edu>; Adrian Self <amself@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

I'll start getting this into Cayuse, @Nicholas Eggleston, will Dan be serving as Co-PI as well, or possibly taking on a scientific advisor role?

--

David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/

☐ Book time to meet with me

From: David Molik <molik@ksu.edu>
Date: Monday, May 4, 2026 at 4:32 PM
To: Nicholas Eggleston <njeggleston@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Gina Becker <gbecker@ksu.edu>; Adrian Self <amself@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

I generally write risks and mitigation strategies directly into my grants.

--

David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange
Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/

☐ Book time to meet with me

From: Nicholas Eggleston <njeggleston@ksu.edu>
Date: Monday, May 4, 2026 at 3:52 PM
To: David Molik <molik@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Gina Becker <gbecker@ksu.edu>; Adrian Self <amself@ksu.edu>
Subject: Re: NSF IDSS Proposal Concept

I definitely endorse this concept. My biggest hangup would probably be ensuring the data center and other infrastructure that underpins Beocat (admin servers, networking storage, etc.) is up to snuff enough to support this endeavor. I'm working on a plan to shore everything up but I just want to call it out as a risk ahead of time. I don't know if we want to write mitigations into this grant or if that would make it too large?

Nick

--

Nick Eggleston
Beocat Team Lead
High Performance Computing Architect
Institute for Computational Research in Engineering and Science

---

From: David Molik <molik@ksu.edu>
Sent: Monday, May 4, 2026 3:32 PM
To: Nicholas Eggleston <njeggleston@ksu.edu>; James Wiser <jawiser@ksu.edu>; Daniel Andresen <dan@ksu.edu>
Cc: Gina Becker <gbecker@ksu.edu>; Adrian Self <amself@ksu.edu>
Subject: NSF IDSS Proposal Concept

Hi Nicholas and Dan,

I took the weekend to think this through, and if there are no objections from CSE or Beocat, I'm planning to move forward with an NSF IDSS proposal, due July 28:

https://www.nsf.gov/funding/opportunities/idss-integrated-data-systems-services

The proposal would aim to build a shared research computing and open research infrastructure layer for K-State, with an emphasis on reproducible, data-intensive, and AI-enabled research workflows. I am framing this less as a general compute acquisition proposal and more as an operations-level data and workflow service, which is where IDSS appears to fit best.

Right now, I am imagining four connected deliverables:

Deliverable 1: Open Reproducible Research Environment Registry
A DOI-linked registry for containers, workflows, AI models, computational environments, metadata, provenance, and citation files.

Deliverable 2: Kubernetes-Native Data/AI Workflow Execution Service
A Beocat-backed operational service that allows registered environments to be executed against research datasets through portable, reproducible workflows.

Deliverable 3: Research Object and Public Access Integration Layer
OPE-led integration connecting software, containers, workflows, datasets, publications, DOI metadata, repository records, and public-access compliance.

Deliverable 4: Operations, Governance, Metrics, and Emerging Architecture Testbed
National onboarding, user support, training, security, performance measures, sustainability planning, and a Sunderland-based evaluation testbed for non-GPU AI architectures.

I am deliberately including the operations, governance, metrics, security/trustworthiness, project management, and sustainability pieces because IDSS Category II needs to be framed around operational readiness and national-scale service capacity, more so than around local infrastructure. The software engineering component is also central here; this is not just a hardware or cluster expansion proposal.

I'm mostly interested in this because of a campus need: the larger goal here is to start building toward the kind of shared research computing infrastructure center that many peer institutions already have: a coordinated research arm, reusable technical capacity, and pooled technical/software engineering hours that can be written into future grants. I am thinking here of models like Notre Dame's Center for Research Computing, where technical staff and infrastructure are part of the research development ecosystem rather than treated only as local service capacity. A software engineering heavy proposal like this is going to go to go a long way towards building those competences.

Nicholas and Dan, I suspect one or both of you may want to be involved as Co-PIs, assuming the concept makes sense from your side. At this stage, I'm trying to pressure-test the Beocat/CSE technical components: the Kubernetes execution layer, software engineering scope, security model, and how best to make the non-GPU AI testbed accessible through Beocat-facing workflows.

Best,
David
--
David Molik (Ph.D.)
Associate Professor and Librarian
Department Head, Open Publishing Exchange

Kansas State University
molik@ksu.edu | (785-532-5399)
lib.k-state.edu/about/our-people/david-molik/

Book time to meet with me